

Data Privacy: Homeworks #4

Due on Monday 11PM, June 8, 2026

Note: Submit homework via LearnUS.

1. **Neyman–Pearson Lemma.** Let T be a real-valued random variable. Under hypothesis H_b (for $b \in \{0, 1\}$), assume T has a continuous density p_b on $\mathbb{R}$, with $p_0(t) > 0$ for every t . Define the *likelihood ratio*

$$\Lambda(t) = \frac{p_1(t)}{p_0(t)}.$$

A *test* is a function $\phi : \mathbb{R} \rightarrow [0, 1]$, where $\phi(t)$ is the probability of rejecting H_0 after observing $T = t$. Define

$$\alpha(\phi) = \mathbb{E}_{H_0}[\phi(T)] \quad (\text{size, type-I error}), \quad \beta(\phi) = \mathbb{E}_{H_1}[\phi(T)] \quad (\text{power}).$$

Fix a level $\alpha_0 \in (0, 1)$, and *assume that the function* $G(\tau) := \Pr_{H_0}[\Lambda(T) > \tau]$ *is continuous on* $[0, \infty)$. Prove the following three claims.

- (a) **Existence.** Show that there exists a threshold $\tau^* \geq 0$ such that the test

$$\phi^*(t) = \mathbf{1}[\Lambda(t) > \tau^*]$$

satisfies $\alpha(\phi^*) = \alpha_0$ exactly.

Hint: Show that G is non-increasing on $[0, \infty)$ with $G(0) \leq 1$ and $G(\tau) \rightarrow 0$ as $\tau \rightarrow \infty$. Then apply the intermediate value theorem.

- (b) **Optimality.** Show that for every test ϕ with $\alpha(\phi) \leq \alpha_0$, we have $\beta(\phi) \leq \beta(\phi^*)$.

Hint: Verify that for every t , $(\phi^*(t) - \phi(t))(p_1(t) - \tau^* p_0(t)) \geq 0$, then integrate over $\mathbb{R}$.

- (c) **Uniqueness.** Suppose ϕ is any most-powerful level- α_0 test (i.e., $\alpha(\phi) \leq \alpha_0$ and $\beta(\phi) = \beta(\phi^*)$). Show that $\phi(t) = \phi^*(t)$ for every t with $\Lambda(t) \neq \tau^*$, except possibly on a set of probability 0.

Hint: The integrand from part (b) is non-negative and now integrates to 0.

2. **DP $\Rightarrow$ MIA Bounds.** Let $M : \mathcal{X}^n \rightarrow \mathcal{Y}$ be (ϵ, δ) -differentially private, and let D_0, D_1 be two neighboring datasets. Consider any (possibly randomized) binary test $\mathcal{A} : \mathcal{Y} \rightarrow \{0, 1\}$ that tries to distinguish $M(D_0)$ from $M(D_1)$. Write

$$\text{FPR}(\mathcal{A}) = \Pr[\mathcal{A}(M(D_0)) = 1], \quad \text{TPR}(\mathcal{A}) = \Pr[\mathcal{A}(M(D_1)) = 1].$$

- (a) Show that

$$\text{TPR}(\mathcal{A}) \leq e^\epsilon \text{FPR}(\mathcal{A}) + \delta, \quad 1 - \text{FPR}(\mathcal{A}) \leq e^\epsilon (1 - \text{TPR}(\mathcal{A})) + \delta.$$

- (b) Define the MI advantage $\text{Adv}(\mathcal{A}) = \text{TPR}(\mathcal{A}) - \text{FPR}(\mathcal{A})$. Show the loose bound

$$\text{Adv}(\mathcal{A}) \leq e^\epsilon - 1 + \delta.$$

Then, by combining both inequalities of part (a), derive the tighter Kairouz–Oh–Viswanath bound

$$\text{Adv}(\mathcal{A}) \leq \frac{e^\epsilon - 1 + 2\delta}{e^\epsilon + 1}.$$

- (c) Conclude that for $\varepsilon = 0.1$ and $\delta = 10^{-5}$, no adversary can exceed advantage ≈ 0.105 , but for $\varepsilon = 8$ the bound is vacuous.

3. **Generalization Gap as a Leakage Lower Bound (Yeom).** Suppose the loss is bounded: $\ell(f_\theta, z) \in [0, B]$ for all θ, z . Define

$$R_{\text{train}} = \mathbb{E}_{z \sim \text{Unif}(D)}[\ell(f_\theta, z)], \quad R_{\text{pop}} = \mathbb{E}_{z \sim \mathcal{P}}[\ell(f_\theta, z)], \quad \Delta = R_{\text{pop}} - R_{\text{train}} \geq 0.$$

The MIA game: sample $b \sim \text{Unif}\{0, 1\}$. If $b = 1$, return z uniform over D ; otherwise, return $z \sim \mathcal{P}$. The adversary uses a threshold attack $\mathcal{A}_\tau(z) = \mathbf{1}[\ell(f_\theta, z) \leq \tau]$. Let $F_1(\tau) = \Pr[\ell \leq \tau \mid b = 1]$ and $F_0(\tau) = \Pr[\ell \leq \tau \mid b = 0]$; assume both have continuous densities on $[0, B]$.

- (a) Show that the advantage of $\mathcal{A}_\tau$ is

$$\text{Adv}(\mathcal{A}_\tau) = F_1(\tau) - F_0(\tau).$$

- (b) Prove the *layer-cake identity*: for any random variable X with continuous density on $[0, B]$,

$$\mathbb{E}[X] = \int_0^B (1 - F_X(t)) dt.$$

Then deduce

$$\int_0^B (F_1(t) - F_0(t)) dt = \Delta.$$

- (c) Conclude that there exists a threshold $\tau^* \in [0, B]$ for which the threshold attack achieves advantage

$$\text{Adv}(\mathcal{A}_{\tau^*}) \geq \frac{\Delta}{B}.$$

In words: the generalization gap is a *lower* bound on the leakage of the best threshold attack — not an upper bound, contrary to a common informal slogan.

Hint: $\Delta = \int_0^B (F_1 - F_0) dt \leq B \cdot \sup_t (F_1(t) - F_0(t))$.

- (d) Now suppose additionally that under each hypothesis the loss is Gaussian:

$$\ell \mid b = 1 \sim \mathcal{N}(R_{\text{train}}, \sigma^2), \quad \ell \mid b = 0 \sim \mathcal{N}(R_{\text{pop}}, \sigma^2).$$

(Ignore the issue that Gaussians are not literally bounded; the formula below is the correct one for the full range $\mathbb{R}$.) Show that the maximum-advantage threshold attack has $\tau^* = (R_{\text{train}} + R_{\text{pop}})/2$ and achieves

$$\text{Adv}^* = 2\Phi\left(\frac{\Delta}{2\sigma}\right) - 1.$$

For small Δ/σ , derive the linear approximation $\text{Adv}^* \approx \Delta/(\sigma\sqrt{2\pi})$ and explain why noisier loss distributions reduce leakage.

4. **Influence-Function Unlearning (Closed Form).** Let $L(\theta; D) = \frac{1}{n} \sum_{i=1}^n \ell(\theta; z_i)$ be a twice-differentiable, λ -strongly convex loss in $\theta \in \mathbb{R}^d$. Recall that λ -strong convexity means the Hessian $H_\theta = \nabla^2 L(\theta; D)$ satisfies $H_\theta \succeq \lambda I$, which in turn implies H_θ is invertible with $\|H_\theta^{-1}\|_{\text{op}} \leq 1/\lambda$. Let $\theta_o = \arg \min_\theta L(\theta; D)$ and $\theta_{-z} = \arg \min_\theta L(\theta; D \setminus \{z\})$.

(a) Consider the parametric family

$$L_t(\theta) = L(\theta; D) - \frac{t}{n} \ell(\theta; z), \quad t \in [0, 1],$$

and let $\theta(t) = \arg \min_\theta L_t(\theta)$. (Note that $L_0 = L(\cdot; D)$ gives $\theta(0) = \theta_o$, and L_1 is a positive scalar multiple of $L(\cdot; D \setminus \{z\})$, so $\theta(1) = \theta_{-z}$.) Use the first-order optimality condition $\nabla_\theta L_t(\theta(t)) = 0$ and differentiate with respect to t (the *implicit function theorem*) to derive

$$\left. \frac{d\theta(t)}{dt} \right|_{t=0} = \frac{1}{n} H_{\theta_o}^{-1} \nabla_\theta \ell(\theta_o; z).$$

(b) Conclude the first-order Newton-step approximation

$$\theta_{-z} - \theta_o \approx \frac{1}{n} H_{\theta_o}^{-1} \nabla_\theta \ell(\theta_o; z).$$

The “unlearned” parameter is then $\theta_u(D, z) = \theta_o + \frac{1}{n} H_{\theta_o}^{-1} \nabla_\theta \ell(\theta_o; z)$.

(c) Assume in addition that ℓ is G -Lipschitz in θ (so $\|\nabla_\theta \ell(\theta; z)\| \leq G$ for all θ, z). Show that the L_2 -sensitivity of the unlearned parameter to the choice of deleted point is bounded:

$$\sup_{z, z'} \|\theta_u(D, z) - \theta_u(D, z')\| \leq \frac{2G}{\lambda n}.$$

(Treat $H_{\theta_o}^{-1}$ as fixed; its operator norm is $\leq 1/\lambda$ by strong convexity. This is the first-order sensitivity used in certified unlearning.)

5. **Certified (ε, δ) -Unlearning by Gaussian Noise.** Building on the previous problem, let $A(D) = \arg \min_\theta L(\theta; D)$ be the (deterministic) full-retrain map, and define the noisy first-order unlearning algorithm

$$\mathcal{U}(D, z) = \theta_o(D) + \frac{1}{n} H_{\theta_o(D)}^{-1} \nabla_\theta \ell(\theta_o(D); z) + N, \quad N \sim \mathcal{N}(0, \sigma^2 I_d).$$

Assume ℓ is λ -strongly convex and G -Lipschitz, and that the first-order approximation in Problem 4(b) has L_2 error bounded by

$$\eta := \|\theta_u^{\text{1st}} - A(D \setminus \{z\})\| \leq \frac{C_2 G^2}{\lambda^2 n^2}$$

for some universal constant C_2 (a known second-order bound that you may use without proof).

- (a) Show that the deterministic part of $\mathcal{U}(D, z)$ differs from $A(D \setminus \{z\})$ in L_2 norm by at most

$$\eta \leq \frac{C_2 G^2}{\lambda^2 n^2}.$$

- (b) Recall the Gaussian-mechanism privacy bound: if $f, g : \mathcal{X}^n \rightarrow \mathbb{R}^d$ satisfy $\|f(X) - g(X)\| \leq \eta$ for every X , then $f(X) + N$ and $g(X) + N$ are (ε, δ) -indistinguishable provided

$$\sigma \geq \frac{\eta}{\varepsilon} \sqrt{2 \log(1.25/\delta)}.$$

Apply this with $f = \theta_u^{\text{1st}}$ and $g = A(\cdot \setminus \{z\})$ to conclude that

$$\sigma \geq \frac{C_2 G^2}{\lambda^2 n^2 \varepsilon} \sqrt{2 \log(1.25/\delta)}$$

yields an (ε, δ) -certified unlearning algorithm.

- (c) How does the required noise scale with n ? Compare to the Gaussian-mechanism noise needed for (ε, δ) -DP of the trained model itself ($\sigma \propto 1/(n\varepsilon)$). Explain in one or two sentences why the unlearning noise is smaller for large n .